1. The Open Systems Interconnection (OSI) reference model developed by ISO consists of how many hierarchical layers?

(A) 7 Layers
(B) 4 Layers
(C) 5 Layers
(D) 6 Layers

2. The correct sequence of the OSI 7 layers from Layer 1 (bottom) to Layer 7 (top) is:

(A) Physical, Data Link, Network, Transport, Session, Presentation, Application
(B) Application, Presentation, Session, Transport, Network, Data Link, Physical
(C) Physical, Network, Data Link, Transport, Session, Presentation, Application
(D) Physical, Data Link, Transport, Network, Session, Presentation, Application

3. The standard TCP/IP protocol suite originally defined in RFC 1122 is structured into how many layers?

(A) 4 (or 5 in modern textbooks)
(B) 7 Layers
(C) 3 Layers
(D) 6 Layers

4. Which layer of the OSI model is responsible for transmitting raw unstructured bit streams over a physical communication medium?

(A) Physical Layer (Layer 1)
(B) Data Link Layer
(C) Network Layer
(D) Transport Layer

5. The Data Link Layer (Layer 2) groups raw bits received from the physical layer into logical data units termed:

(A) Frames
(B) Packets
(C) Segments
(D) Datagrams

6. The Protocol Data Unit (PDU) at the Network Layer (Layer 3) is commonly known as a:

(A) Packet (or Datagram)
(B) Frame
(C) Segment
(D) Bit stream

7. The Protocol Data Unit (PDU) at the Transport Layer (Layer 4) in TCP is designated as a:

(A) Segment
(B) Packet
(C) Frame
(D) Message

8. The process of prepending protocol header information as data moves DOWN the protocol stack from Layer 7 to Layer 1 is:

(A) Encapsulation
(B) Decapsulation
(C) Multiplexing
(D) Demodulation

9. The reverse process of stripping off protocol headers as data moves UP the receiving protocol stack is:

(A) Decapsulation
(B) Encapsulation
(C) Modulation
(D) Segmentation

10. Which OSI layer is responsible for end-to-end (process-to-process) reliable communication and error recovery?

(A) Transport Layer
(B) Network Layer
(C) Data Link Layer
(D) Session Layer

11. Which OSI layer provides logical addressing (IP addressing) and host-to-host path determination (routing)?

(A) Network Layer
(B) Data Link Layer
(C) Transport Layer
(D) Physical Layer

12. Which OSI layer handles data translation, ASCII/Unicode encoding, data compression, and encryption/decryption?

(A) Presentation Layer (Layer 6)
(B) Application Layer
(C) Session Layer
(D) Transport Layer

13. Which OSI layer manages dialogue control, session checkpointing, and token synchronization between endpoints?

(A) Session Layer (Layer 5)
(B) Transport Layer
(C) Network Layer
(D) Presentation Layer

14. Which OSI layer interfaces directly with end-user software applications (HTTP, SMTP, FTP, DNS)?

(A) Application Layer (Layer 7)
(B) Presentation Layer
(C) Session Layer
(D) Transport Layer

15. In guided transmission media, signals are propagated along a physical solid conduit. Which is an example of guided media?

(A) Twisted-Pair Copper Cable, Coaxial Cable, and Fiber-Optic Cable
(B) Satellite microwave
(C) Terrestrial radio waves
(D) Infrared free space

16. Why are the two insulated copper conductors in a Twisted-Pair cable physically twisted around each other?

(A) To cancel out electromagnetic interference (EMI) and reduce crosstalk from adjacent pairs
(B) To increase physical tensile strength
(C) To increase electrical resistance
(D) To lower the speed of light

17. What is the primary physical distinction between Unshielded Twisted Pair (UTP) and Shielded Twisted Pair (STP)?

(A) STP incorporates a metallic foil/braided shield around wire pairs to block severe external electromagnetic noise
(B) UTP has thicker copper wires
(C) STP cannot support Gigabit Ethernet
(D) UTP requires grounding at both ends

18. Which twisted-pair cable category is widely deployed for Gigabit Ethernet (1000BASE-T) supporting frequencies up to 100 MHz?

(A) Category 5e (Cat 5e)
(B) Category 3
(C) Category 1
(D) Category 2

19. Coaxial cable consists of a central copper core surrounded by an insulator, a braided metal shield, and an outer jacket, commonly used for:

(A) Cable television (CATV), broadband Internet, and legacy 10BASE2/10BASE5 Ethernet
(B) Telephone subscriber loops
(C) Long-haul transoceanic fiber links
(D) Sub-nanosecond chip interconnects

20. Fiber-optic cables transmit digital information encoded as pulses of light exploiting the optical principle of:

(A) Total Internal Reflection (TIR)
(B) Refraction only
(C) Diffraction
(D) Polarization

21. For total internal reflection to occur in an optical fiber, the refractive index of the core (n1) must be:

(A) Strictly greater than the refractive index of the cladding (n1 > n2)
(B) Strictly less than the cladding (n1 < n2)
(C) Equal to the cladding (n1 = n2)
(D) Zero

22. Single-Mode Fiber (SMF) possesses an extremely narrow core diameter (~8 to 10 micrometers), which provides the advantage of:

(A) Eliminating modal dispersion, enabling transmission over long distances exceeding 50-100 km
(B) Lower optical cable cost
(C) Allowing inexpensive LED light sources
(D) Supporting higher mechanical bending

23. Multi-Mode Fiber (MMF) has a larger core diameter (~50 to 62.5 micrometers), but its maximum transmission distance is limited by:

(A) Modal Dispersion (different light rays travelling along different optical paths)
(B) High electrical resistance
(C) Thermal noise
(D) Magnetic saturation

24. Unguided transmission media transport electromagnetic waves without a physical conduit, collectively referred to as:

(A) Wireless Communications
(B) Guided conduits
(C) Coaxial lines
(D) Metallic wave guides

25. Radio waves typically occupy the electromagnetic frequency spectrum from:

(A) 3 kHz to 1 GHz
(B) 1 GHz to 300 GHz
(C) 300 GHz to 400 THz
(D) 400 THz to 800 THz

26. Microwaves occupy the electromagnetic spectrum ranging from 1 GHz to 300 GHz, exhibiting the key propagation property of:

(A) Line-of-Sight (LOS) directional beam propagation
(B) Omnidirectional ground wave bending
(C) Ionospheric sky wave bouncing
(D) Penetrating thick mountains

27. Infrared wireless signals (300 GHz to 400 THz) used in television remote controls cannot penetrate solid walls, which provides:

(A) Inherent security against neighboring eavesdropping within an enclosed room
(B) Extremely long transmission ranges
(C) High atmospheric penetration in rain
(D) Omnidirectional multi-room coverage

28. The loss of signal energy and amplitude as it propagates over distance along a transmission medium is termed:

(A) Attenuation
(B) Distortion
(C) Dispersion
(D) Crosstalk

29. To compensate for signal attenuation over long copper cables, which hardware device is inserted to restore signal strength?

(A) Amplifier (for analog signals) or Repeater (for digital bit streams)
(B) Modem
(C) Switch
(D) Attenuator

30. Signal attenuation in telecommunications is mathematically measured in logarithmic units called:

(A) Decibels (dB): dB = 10 * log10(P2 / P1)
(B) Volts
(C) Amperes
(D) Ohms

31. The phenomenon where a signal alters its waveform shape because different composite frequency components travel at different velocities is:

(A) Distortion (Delay Distortion)
(B) Attenuation
(C) Thermal noise
(D) Crosstalk

32. Thermal noise (Johnson-Nyquist noise) present in all electronic communication circuits is generated by:

(A) Random thermal agitation of free electrons inside the conductive medium
(B) Lightning strikes
(C) Adjacent phone lines
(D) Loose cable connectors

33. The unwanted inductive or capacitive coupling of signals from one adjacent wire pair into another is termed:

(A) Crosstalk
(B) Impulse noise
(C) White noise
(D) Aliasing

34. The Signal-to-Noise Ratio (SNR) is mathematically defined as the ratio of:

(A) Average Signal Power to Average Noise Power: SNR = P_signal / P_noise
(B) Noise Power to Signal Power
(C) Peak Voltage to RMS Voltage
(D) Bandwidth to Frequency

35. In telecommunications, an SNR value of 1000 converted to the logarithmic decibel scale (SNR_dB) equals:

(A) 30 dB (10 * log10(1000) = 10 * 3)
(B) 10 dB
(C) 20 dB
(D) 3 dB

36. A digital signal transmission method where bits are sent directly over the medium without modulation using their original frequency spectrum is:

(A) Baseband Transmission
(B) Broadband Transmission
(C) Passband Transmission
(D) Frequency Hopping

37. Broadband Transmission modulates a digital signal onto a high-frequency analog sinusoidal carrier wave using a:

(A) Modem (Modulator-Demodulator)
(B) Repeater
(C) Switch
(D) Hub

38. In digital modulation, varying the amplitude of the carrier sine wave to represent binary 0 and 1 is termed:

(A) Amplitude Shift Keying (ASK)
(B) Frequency Shift Keying (FSK)
(C) Phase Shift Keying (PSK)
(D) Quadrature Modulation

39. In Frequency Shift Keying (FSK), binary 0 and 1 are represented by modulating the carrier's:

(A) Frequency (two distinct frequencies f1 and f2)
(B) Phase angle
(C) Amplitude
(D) Polarization

40. In Phase Shift Keying (PSK), binary data is encoded by altering the carrier's:

(A) Phase angle (e.g., 0 degrees for bit 1, 180 degrees for bit 0 in BPSK)
(B) Amplitude
(C) Frequency
(D) Wavelength

41. Quadrature Amplitude Modulation (QAM) achieves high spectral efficiency by simultaneously modulating both:

(A) Amplitude and Phase of two orthogonal carrier waves (In-phase and Quadrature)
(B) Frequency and Wavelength
(C) Time and Duty cycle
(D) Voltage and Resistance

42. How many bits per symbol are transmitted in a 16-QAM digital constellation scheme?

(A) 4 bits per symbol (2^4 = 16)
(B) 2 bits per symbol
(C) 8 bits per symbol
(D) 16 bits per symbol

43. In a 64-QAM digital constellation modulation scheme, each transmitted baud symbol represents:

(A) 6 bits (2^6 = 64)
(B) 8 bits
(C) 4 bits
(D) 12 bits

44. According to Nyquist's theorem, what is the maximum theoretical bitrate over a noiseless 4 kHz channel using 8 discrete signal levels?

(A) 24,000 bps (2 * 4000 * log2(8) = 8000 * 3)
(B) 12,000 bps
(C) 32,000 bps
(D) 48,000 bps

45. A noiseless channel has bandwidth B = 20 kHz. If each signal element carries 4 bits of information, the Nyquist bitrate limit is:

(A) 160,000 bps (160 kbps)
(B) 80,000 bps
(C) 40,000 bps
(D) 320,000 bps

46. Shannon's capacity formula demonstrates that to achieve non-zero channel capacity across an extremely noisy channel, one must:

(A) Increase channel bandwidth B or boost signal transmission power
(B) Decrease signal power
(C) Eliminate all amplifiers
(D) Reduce sampling frequency

47. A fiber optic communication channel has bandwidth B = 10 MHz and SNR = 15. The theoretical Shannon capacity is:

(A) 40 Mbps (10 * log2(1 + 15) = 10 * 4)
(B) 20 Mbps
(C) 80 Mbps
(D) 150 Mbps

48. Pulse Code Modulation (PCM) converts continuous analog waveforms into digital bit streams through three sequential stages:

(A) Sampling, Quantization, and Binary Encoding
(B) Modulation, Amplification, and Detection
(C) Filtering, Rectification, and Smoothing
(D) Compression, Encryption, and Framing

49. In PCM speech transmission, the analog voice spectrum is bandlimited to 4 kHz. The minimum Nyquist sampling rate required is:

(A) 8,000 samples per second (8 kHz)
(B) 4,000 samples per second
(C) 16,000 samples per second
(D) 2,000 samples per second

50. If each voice sample in Q49 is quantized into 8 bits (256 discrete levels), the standard PCM voice bit rate is:

(A) 64 kbps (8,000 samples/s * 8 bits/sample = DSO channel)
(B) 32 kbps
(C) 128 kbps
(D) 56 kbps

51. The difference between the actual continuous analog sample voltage and the discrete quantized approximation level is termed:

(A) Quantization Error (Quantization Noise)
(B) Thermal noise
(C) Harmonic distortion
(D) Jitter

52. In Non-Return-to-Zero-Level (NRZ-L) digital line encoding, a positive voltage typically represents binary 0 and negative voltage represents 1. Its main drawback is:

(A) Lack of self-clocking synchronization during long consecutive sequences of identical bits and DC baseline wander
(B) High complexity
(C) Excessive bandwidth consumption
(D) High manufacturing cost

53. Manchester line encoding (used in legacy 10 Mbps Ethernet 802.3) guarantees self-clocking synchronization by:

(A) Mandating a voltage transition in the exact middle of EVERY bit interval
(B) Doubling the signal amplitude
(C) Inverting voltage every second bit
(D) Inserting stop bits

54. In Manchester encoding, a binary 1 is conventionally represented by:

(A) A Low-to-High transition at the bit center
(B) A High-to-Low transition at bit center
(C) Constant high voltage
(D) Zero voltage

55. Differential Manchester encoding (used in Token Ring IEEE 802.5) indicates binary data by:

(A) The presence or absence of a transition at the START of the bit interval (transition for 0, no transition for 1)
(B) The absolute voltage polarity
(C) A pulse width variation
(D) Carrier phase shifts

56. Bipolar AMI (Alternate Mark Inversion) line coding represents binary 0 as zero voltage and binary 1 as:

(A) Alternating positive and negative voltage pulses, eliminating DC component
(B) Constant positive voltage
(C) High frequency tone
(D) Half-cycle sine wave

57. B8ZS (Bipolar with 8-Zero Substitution) scrambling used in North American T1 carrier lines prevents sync loss by:

(A) Substituting a sequence of eight consecutive zeros with an intentional bipolar violation pattern
(B) Inserting parity bits every 4 bits
(C) Compressing the payload
(D) Adding stop bits

58. Multiplexing is the telecommunication technique that combines multiple independent data signals for simultaneous transmission over a:

(A) Single shared high-capacity physical link
(B) Dedicated private cable per user
(C) Local loop switch
(D) Modem bank

59. Frequency-Division Multiplexing (FDM) is an analog multiplexing technique where:

(A) The total channel bandwidth is divided into non-overlapping frequency bands separated by guard bands
(B) Time is divided into discrete slots
(C) Each user gets a unique pseudorandom code
(D) Light wavelengths are polarized

60. Wavelength-Division Multiplexing (WDM) applied in fiber-optic communications is essentially:

(A) Optical FDM, transmitting multiple laser light beams of different wavelengths (colors) down a single optical fiber
(B) Time-slicing an LED source
(C) Pulse code modulation of fiber
(D) Electrical baseband switching

61. Dense Wavelength-Division Multiplexing (DWDM) packs optical channels tightly with channel spacing as narrow as:

(A) 0.8 nm or 0.4 nm (100 GHz or 50 GHz frequency spacing), carrying over 80-160 optical channels
(B) 20 nm
(C) 100 nm
(D) 500 nm

62. Synchronous Time-Division Multiplexing (TDM) allocates time slots to connected data sources such that:

(A) Each input device is assigned a fixed, pre-allocated time slot in every recurring TDM frame, even if it has no data to send
(B) Time slots are allocated dynamically on demand
(C) Slots are assigned by random contention
(D) High priority packets preempt lower slots

63. Statistical (Asynchronous) TDM improves on Synchronous TDM by:

(A) Dynamically allocating time slots only to active channels that currently possess data, eliminating wasted idle slots
(B) Using fixed frequency bands
(C) Requiring analog modulation
(D) Operating without buffers

64. A standard North American Digital Signal 1 (DS1) / T1 carrier frame aggregates 24 voice channels using TDM. Each T1 frame consists of:

(A) 193 bits (24 channels * 8 bits/channel + 1 framing bit), operating at 1.544 Mbps
(B) 128 bits at 1.0 Mbps
(C) 256 bits at 2.048 Mbps
(D) 192 bits at 1.536 Mbps

65. The European E1 carrier standard multiplexes 32 digital channels (30 voice channels + 1 framing + 1 signaling), operating at:

(A) 2.048 Mbps (32 * 8 bits * 8000 frames/s)
(B) 1.544 Mbps
(C) 64 kbps
(D) 4.096 Mbps

66. The optical signal transmission loss through modern silica single-mode fiber is minimized in which infrared telecommunication wavelength window?

(A) 1550 nm window (~ 0.2 dB/km attenuation)
(B) 850 nm window
(C) 1310 nm window (~ 0.35 dB/km)
(D) 650 nm visible red window

67. Chromatic Dispersion in optical fiber communication occurs because:

(A) Different spectral wavelengths of light emitted by the laser source travel at slightly different group velocities through silica glass
(B) The fiber core is bent physically
(C) Light reflects off the outer buffer jacket
(D) Core diameter fluctuates randomly

68. Polarization Mode Dispersion (PMD) arises in long-haul optical fibers due to:

(A) Slight geometric non-circularity (asymmetry) of the fiber core causing two orthogonal polarization states to travel at different speeds
(B) Excessive laser optical power
(C) Microbending attenuation
(D) Photodiode saturation

69. Erbium-Doped Fiber Amplifiers (EDFA) revolutionized long-distance optical networking because they:

(A) Amplify optical signals directly in the optical domain around 1550 nm without requiring costly optical-electronic-optical (OEO) conversion
(B) Double the core diameter
(C) Filter out high frequencies
(D) Convert light into microwave signals

70. What is the typical velocity factor of electrical signals propagating through standard copper twisted-pair cable?

(A) Approximately 0.67c (~ 2.0 * 10^8 m/s, where c is speed of light in vacuum)
(B) 1.0c (3.0 * 10^8 m/s)
(C) 0.1c (3.0 * 10^7 m/s)
(D) 0.99c

71. The characteristic impedance of standard Category 5e/6 UTP network cabling is:

(A) 100 Ohms (+/- 15 Ohms)
(B) 50 Ohms
(C) 75 Ohms
(D) 300 Ohms

72. The characteristic impedance of standard RG-6 coaxial cable used for cable television and cable modems is:

(A) 75 Ohms
(B) 50 Ohms
(C) 100 Ohms
(D) 120 Ohms

73. Return Loss in transmission lines measures the ratio of incident power to reflected power in decibels, caused by:

(A) Impedance mismatches at connectors, terminations, or structural discontinuities
(B) DC cable resistance
(C) High environmental humidity
(D) Electromagnetic radiation loss

74. Near-End Crosstalk (NEXT) measures signal interference coupled into an adjacent wire pair measured at:

(A) The same end of the cable link where the interfering transmitter is located
(B) The opposite receiving end of the cable
(C) The exact midpoint of the cable
(D) The central switch port

75. Power Sum NEXT (PS-NEXT) is critical for Gigabit Ethernet (1000BASE-T) because:

(A) All 4 wire pairs transmit and receive simultaneously, so each pair receives crosstalk from the other 3 active pairs simultaneously
(B) Only one pair is active at a time
(C) It uses fiber optic transceivers
(D) Power is transmitted via PoE

76. A digital communication signal with input power P1 = 100 mW is transmitted over an optical fiber link with total attenuation of 15 dB. The output power P2 is:

(A) 3.16 mW (P2 = P1 * 10^(- 15/10) = 100 * 0.03162)
(B) 1.00 mW
(C) 10.0 mW
(D) 0.316 mW

77. If a signal with initial power of 2 mW is amplified by an amplifier with gain + 20 dB and then passes through a cable with loss - 14 dB, the final power is:

(A) 7.96 mW (Net gain = + 6 dB => Power factor = 10^0.6 ~= 3.98 => 2 * 3.98)
(B) 4.00 mW
(C) 12.5 mW
(D) 1.58 mW

78. Calculate the theoretical Shannon capacity for a channel with bandwidth B = 1 MHz and an SNR of 63 (approx 18 dB):

(A) 6.0 Mbps (1 * 10^6 * log2(1 + 63) = 10^6 * 6)
(B) 3.0 Mbps
(C) 12.0 Mbps
(D) 18.0 Mbps

79. To transmit data at 56 kbps over a telephone channel of bandwidth B = 3.4 kHz, what minimum SNR (in dB) is theoretically required by Shannon's theorem?

(A) 49.5 dB (56000 / 3400 = log2(1 + SNR) => 16.47 bits/Hz => SNR = 2^16.47 ~= 90,000 => 49.5 dB)
(B) 30.0 dB
(C) 20.5 dB
(D) 60.0 dB

80. In a Constellation Diagram for digital modulation, what do the radial distance from origin and the angle from positive x-axis represent?

(A) Carrier Amplitude (magnitude) and Carrier Phase angle
(B) Frequency and Wavelength
(C) Signal-to-Noise ratio and Bitrate
(D) Baud rate and Group delay

81. A digital communication system operates at a baud rate (modulation rate) of 2400 symbols per second. If 256-QAM modulation is used, the data bitrate is:

(A) 19,200 bps (2400 * log2(256) = 2400 * 8)
(B) 9,600 bps
(C) 38,400 bps
(D) 4,800 bps

82. Bit Error Rate (BER) is mathematically defined as:

(A) The ratio of the number of erroneously received bits to the total number of transmitted bits over a given time interval
(B) The number of dropped packets per minute
(C) The ratio of bandwidth to latency
(D) The percentage of retransmitted frames

83. In fiber optic communications, the Numerical Aperture (NA) measures the light-gathering capability of the fiber, given by:

(A) NA = sqrt(n_core^2 - n_cladding^2) = sin(theta_acceptance)
(B) NA = n_core / n_cladding
(C) NA = n_core * n_cladding
(D) NA = sqrt(n_core + n_cladding)

84. An optical fiber has core index n1 = 1.48 and cladding index n2 = 1.46. The Numerical Aperture (NA) and acceptance angle theta_acc are:

(A) NA = sqrt(1.48^2 - 1.46^2) = sqrt(0.0588) = 0.2425; theta_acc = asin(0.2425) = 14.03 degrees
(B) NA = 0.450; theta_acc = 26.7 deg
(C) NA = 0.120; theta_acc = 6.9 deg
(D) NA = 0.020; theta_acc = 1.15 deg

85. The critical angle theta_c for total internal reflection inside the fiber core of Q84 (n1 = 1.48, n2 = 1.46) is:

(A) theta_c = asin(n2 / n1) = asin(1.46 / 1.48) = asin(0.98648) = 80.57 degrees
(B) 60.25 degrees
(C) 45.00 degrees
(D) 89.20 degrees

86. In Time-Division Multiplexing of 10 digital sources each transmitting at 100 kbps, if each source is allocated 1 byte per slot, the aggregate output frame rate is:

(A) 12,500 frames per second (100,000 bits/s / 8 bits = 12,500 bytes/s)
(B) 10,000 frames/s
(C) 100,000 frames/s
(D) 1,250 frames/s

87. For the TDM system in Q86, the total aggregate output link transmission bitrate (excluding framing overhead) is:

(A) 1.0 Mbps (10 sources * 100 kbps)
(B) 800 kbps
(C) 1.25 Mbps
(D) 10 Mbps

88. In Frequency-Hopping Spread Spectrum (FHSS), security and jam-resistance are achieved by:

(A) Rapidly switching the carrier frequency across a wide pseudo-random sequence of frequency channels known only to sender and receiver
(B) Transmitting at maximum possible RF power
(C) Narrowing the channel bandwidth to 1 kHz
(D) Encrypting the cable jacket

89. Direct-Sequence Spread Spectrum (DSSS) modulates each data bit by multiplying it with a high-rate pseudorandom binary code termed a:

(A) Chipping Code (Chip sequence)
(B) Parity vector
(C) Baud sequence
(D) Hamming polynomial

90. Code Division Multiple Access (CDMA) allows multiple users to share the exact same frequency band simultaneously by assigning each user an:

(A) Orthogonal pseudorandom spreading code with zero cross-correlation: Sum(Ci * Cj) = 0 for i != j
(B) Exclusive time slot
(C) Unique carrier frequency
(D) Distinct polarization angle

91. In Orthogonal Frequency Division Multiplexing (OFDM), subcarriers are packed tightly without inter-carrier interference because:

(A) The subcarrier frequency spacing Delta f = 1 / T_symbol ensures mathematical orthogonality over symbol duration T
(B) Guard bands between subcarriers are 10 MHz wide
(C) Subcarriers use different amplitudes
(D) Subcarriers transmit sequentially

92. The Cyclic Prefix (CP) inserted at the beginning of each OFDM symbol serves to:

(A) Act as a guard interval eliminating Inter-Symbol Interference (ISI) caused by multipath channel delay spread
(B) Double the transmission data rate
(C) Provide forward error correction
(D) Synchronize the CPU clock

93. The Free-Space Path Loss (FSPL) for an unguided electromagnetic wave of frequency f propagating over distance d follows the Friis transmission equation:

(A) FSPL = (4 * pi * d * f / c)^2 = (4 * pi * d / lambda)^2
(B) FSPL = 4 * pi * d * f / c
(C) FSPL = d / (4 * pi * f)
(D) FSPL = c^2 / (d * f)

94. Expressed in decibels, the Friis Free-Space Path Loss equation is formulated as:

(A) FSPL(dB) = 20 * log10(d) + 20 * log10(f) + 20 * log10(4*pi / c) = 20*log10(d) + 20*log10(f) - 147.55 dB
(B) FSPL(dB) = 10 * log10(d) + 10 * log10(f)
(C) FSPL(dB) = 40 * log10(d) - 20 * log10(f)
(D) FSPL(dB) = 20 * log10(d / f)

95. Multipath Fading in terrestrial wireless communications occurs due to:

(A) Constructive and destructive interference between multiple radio waves arriving at the receiver antenna via reflection, diffraction, and scattering
(B) Solar flare radiation
(C) Antenna cable resistance
(D) Atmospheric oxygen absorption

96. Rayleigh Fading distribution models the statistical amplitude envelope of a received wireless multipath signal when:

(A) There is NO dominant Line-of-Sight (LOS) path between transmitter and receiver (non-LOS multipath scatter)
(B) A strong direct line-of-sight path exists
(C) The transmitter is stationary and receiver is shielded
(D) Signal frequency is below 1 kHz

97. Rician Fading distribution is used to model multipath wireless propagation when:

(A) A strong direct Line-of-Sight (LOS) path is present alongside weaker scattered multipath components
(B) Zero line-of-sight signal exists
(C) Signal propagates underwater
(D) Transmitter power approaches zero

98. MIMO (Multiple-Input Multiple-Output) antenna technology achieves multiplexing gain by:

(A) Transmitting independent data streams simultaneously over multiple spatial propagation channels using antenna arrays at both ends
(B) Increasing antenna physical size
(C) Using higher RF transmitter wattage
(D) Narrowing the antenna beamwidth

99. Power over Ethernet (PoE, IEEE 802.3bt Type 4) enables twisted-pair Ethernet cables to supply up to what maximum DC power to connected devices?

(A) Up to 90 Watts of DC power alongside Gigabit data
(B) 15 Watts
(C) 500 Watts
(D) 5 Watts

100. In fiber optic communication, Four-Wave Mixing (FWM) is a non-linear Kerr effect that occurs when:

(A) Three optical wavelengths interact in the fiber core to generate a fourth parasitic wavelength, creating severe crosstalk in DWDM systems
(B) Light bounces out of the cladding
(C) Fiber core temperature rises
(D) Photodiode generates dark current

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the layered architecture of the OSI 7-Layer Reference Model. Detail the functions, services, and representative protocol standards associated with each of the seven layers.

Q2. Compare and contrast the OSI Reference Model with the TCP/IP Protocol Suite. Detail their architectural philosophies, layer mappings, and why TCP/IP prevailed in global commercial adoption.

Q3. Formulate and compare the physical characteristics, transmission mechanisms, bandwidth capabilities, and structural shielding of Guided Transmission Media: Twisted Pair, Coaxial Cable, and Fiber Optic Cable.

Q4. State and derive Nyquist's Maximum Bitrate Formula for noiseless channels and Shannon's Channel Capacity Theorem for noisy channels. A communications link has bandwidth B = 4 kHz and Signal-to-Noise Ratio SNR = 3162. Compute the theoretical Shannon capacity.

Q5. Explain Pulse Code Modulation (PCM). Detail the three sequential stages: Sampling, Quantization, and Encoding. Explain the Nyquist sampling criterion and calculate the standard transmission bitrate for a telephone PCM voice channel.

Q6. Analyze physical layer and network modeling topic #6: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q7. Analyze physical layer and network modeling topic #7: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q8. Analyze physical layer and network modeling topic #8: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q9. Analyze physical layer and network modeling topic #9: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q10. Analyze physical layer and network modeling topic #10: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q11. Analyze physical layer and network modeling topic #11: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q12. Analyze physical layer and network modeling topic #12: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q13. Analyze physical layer and network modeling topic #13: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q14. Analyze physical layer and network modeling topic #14: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q15. Analyze physical layer and network modeling topic #15: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q16. Analyze physical layer and network modeling topic #16: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q17. Analyze physical layer and network modeling topic #17: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q18. Analyze physical layer and network modeling topic #18: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q19. Analyze physical layer and network modeling topic #19: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q20. Analyze physical layer and network modeling topic #20: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q21. Analyze physical layer and network modeling topic #21: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q22. Analyze physical layer and network modeling topic #22: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q23. Analyze physical layer and network modeling topic #23: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q24. Analyze physical layer and network modeling topic #24: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q25. Analyze physical layer and network modeling topic #25: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q26. Analyze physical layer and network modeling topic #26: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q27. Analyze physical layer and network modeling topic #27: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q28. Analyze physical layer and network modeling topic #28: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q29. Analyze physical layer and network modeling topic #29: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

Q30. Analyze physical layer and network modeling topic #30: Discuss electromagnetic wave propagation, digital line coding, channel multiplexing, and hardware interfacing.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

comprehensive theoretical model BEFORE protocols were implemented.
Strictly distinguishes between Services, Interfaces, and Protocols. Clean 7-layer theoretical separation.
- TCP/IP Suite: Developed bottom-up by DARPA/ARPA researchers with a pragmatic 'rough consensus and running code' philosophy.
Designed specifically for military survivability and internetworking across disparate networks; protocols were created first, and the model was documented later.

2. Layer-by-Layer Mapping:
- OSI Layers 7, 6, 5 (Application, Presentation, Session) -> Mapped into a single unified TCP/IP Application Layer.
- OSI Layer 4 (Transport) -> Corresponds directly to TCP/IP Transport Layer (Host-to-Host Layer: TCP, UDP).
- OSI Layer 3 (Network) -> Corresponds directly to TCP/IP Internet Layer (IP, ICMP, ARP).
- OSI Layers 2, 1 (Data Link, Physical) -> Mapped into TCP/IP Network Access Layer (or Link Layer / Host-to-Network Layer).

3. Decisive Reasons for TCP/IP's Global Victory:
- 1. Pragmatic Implementation & Timing: TCP/IP code was bundled freely with BSD UNIX (4.2BSD in 1983), providing universities and vendors with an instant, robust, working protocol stack.
- 2. Bureaucratic Overhead of OSI: OSI standard specifications were enormous, overly complex, and bogged down in committee debates for years.
- 3. Redundancy in OSI: OSI implemented flow control and error control redundantly across multiple layers (Layer 2, Layer 3, and Layer 4), increasing software overhead.
- 4. End-to-End Design Principle: TCP/IP keeps the network core simple and stateless (smart endpoints, dumb network), scaling gracefully to billions of nodes.

>>Solution to Question 3: Formulate and compare the physical characteristics, transmission mechanisms, bandwidth capabilities, and structural shielding of Guid

slightly lower refractive index, protected by primary polymer buffer and Kevlar strength yarns.
- Transmission Mechanism: Guided light pulses undergoing Total Internal Reflection (TIR).
- Performance: Virtually limitless bandwidth (tens of Terabits/s via DWDM); attenuation as low as 0.2 dB/km at 1550 nm; transmission distances exceed 100 km without repeaters.
- Immunity: Complete and absolute immunity to electromagnetic interference (EMI), radio frequency interference (RFI), lightning surges, and electrical ground loops; zero crosstalk; highly secure against signal tapping.

= 1 / f_s.
* Nyquist Criterion: To prevent aliasing (spectral overlap), the sampling rate f_s must be at least twice the highest frequency component B present in the signal: f_s >= 2 * B.
- Stage 2: Quantization:
* The infinite continuous voltage amplitudes of the PAM samples are mapped onto a finite set of L discrete, predetermined quantization levels.
* If step size is Delta, the maximum quantization error is e_max = Delta / 2.
* In voice communications, non-uniform logarithmic companding (mu-law in North America, A-law in Europe) provides finer quantization steps for quiet sounds and coarser steps for loud sounds, maintaining constant SNR.
- Stage 3: Binary Encoding:
* Each quantized amplitude level is converted into an n-bit binary code word, where L = 2^n => n = log2(L) bits per sample.

3. Standard Digital Telephone Voice Channel Calculation (DS0 Channel):
- Analog human voice bandwidth is bandlimited by low-pass anti-aliasing filters to B = 4 kHz (300 Hz to 3400 Hz standard speech band).
- Nyquist Sampling Rate: f_s = 2 * B = 2 * 4,000 = 8,000 samples per second.
- Quantization Resolution: n = 8 bits per sample (L = 256 discrete quantization levels).
- Digital Bitrate: Bitrate = f_s * n = 8,000 samples/s * 8 bits/sample = 64,000 bps = 64 kbps.
- This fundamental 64 kbps stream defines the standard International Digital Signal Level 0 (DS0) voice channel.

>>Solution to Question 6: Analyze physical layer and network modeling topic #6: Discuss electromagnetic wave propagation, digital line coding, channel multiplex

3. Practical Network Design Considerations:
Integration into structured cabling standards (TIA/EIA-568), fiber-to-the-home (FTTH) architectures, and wireless 5G cellular physical links.

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3. Practical Network Design Considerations:
Integration into structured cabling standards (TIA/EIA-568), fiber-to-the-home (FTTH) architectures, and wireless 5G cellular physical links.

>>Solution Statement: Analyze physical layer and network modeling topic #13: Discuss electromagnetic wave propagation, digital line coding, channel multiple

3. Practical Network Design Considerations:
Integration into structured cabling standards (TIA/EIA-568), fiber-to-the-home (FTTH) architectures, and wireless 5G cellular physical links.

>>Solution Statement: Analyze physical layer and network modeling topic #15: Discuss electromagnetic wave propagation, digital line coding, channel multiple

3. Practical Network Design Considerations:
Integration into structured cabling standards (TIA/EIA-568), fiber-to-the-home (FTTH) architectures, and wireless 5G cellular physical links.

>>Solution Statement: Analyze physical layer and network modeling topic #19: Discuss electromagnetic wave propagation, digital line coding, channel multiple

3. Practical Network Design Considerations:
Integration into structured cabling standards (TIA/EIA-568), fiber-to-the-home (FTTH) architectures, and wireless 5G cellular physical links.

>>Solution Statement for 20: Analyze physical layer and network modeling topic #20: Discuss electromagnetic wave propagation, digital line coding, channel multiple

3. Practical Network Design Considerations:
Integration into structured cabling standards (TIA/EIA-568), fiber-to-the-home (FTTH) architectures, and wireless 5G cellular physical links.

>>Solution Statement for 28: Analyze physical layer and network modeling topic #28: Discuss electromagnetic wave propagation, digital line coding, channel multiple

3. Practical Network Design Considerations:
Integration into structured cabling standards (TIA/EIA-568), fiber-to-the-home (FTTH) architectures, and wireless 5G cellular physical links.

>>Solution Statement: Analyze physical layer and network modeling topic #26: Discuss electromagnetic wave propagation, digital line coding, channel multiple

3. Practical Network Design Considerations:
Integration into structured cabling standards (TIA/EIA-568), fiber-to-the-home (FTTH) architectures, and wireless 5G cellular physical links.

>>Solution Statement: Analyze physical layer and network modeling topic #20: Discuss electromagnetic wave propagation, digital line coding, channel multiple